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Summary

Hands-on **applied-AI engineering leader** who ships AI products that turn frontier model capabilities into applications people actually use. Currently Head of ML Engineering at Atlassian, the single point of accountability for an extended org of **115+** across Jira Search & Agent, Knowledge Fundamentals, and Generative AI Search Experiences. Previously Group Senior Principal Applied Sciences Manager at Microsoft, built and led a 40+ team that took Bing Question-Answering and Copilot Search from zero to 100+ languages and 230+ markets, and shipped Copilot Search **one month before Google's AI Mode**.

A repeat **0→1 builder —** forms teams, defines products, and ships first versions in ambiguous domains. Operates end-to-end across 0→1 product definition, LLM post-training & distillation, agentic system design, eval methodology, production inference, and enterprise delivery. Uniquely multi-dimensional **— applied sciences & full-stack engineering**, **AI products & research** (50+ publications, 11K+ citations, 20+ patents), **consumer & enterprise products**. 13+ years engineering, 7+ years engineering management.

Selected Highlights — Aligned to Applied AI Leadership

- **Shipping AI products at scale.** A decade of turning frontier model capability into products used by millions — **Atlassian Enterprise Generative AI Search** (0→GA), **Jira Agents in Rovo Chat**, **Bing Question-Answering** (0→global), **Copilot Search** (shipped before Google AI Mode), and **Knowledge Card** (65M DAU).
- **Repeat 0→1 builder.** Built Bing QnA from zero; **Jira Search MVP in 4 months** → 3.4× MAU; **Enterprise Generative AI Search —** identified a gap no team owned, POC in <1 month, GA early 2026; established greenfield sites in Canada and Singapore.
- **LLM application & agentic systems.** Built **Jira Agents in Rovo Chat** (beat Manus/GenSpark on GAIA); pioneered **X-SuperAgent** (first OpenClaw-compliant agent framework at Atlassian); end-to-end LLM post-training & SLM distillation for production serving **— Glow** cut first-token latency from 20–30 s to ∼5 ms.
- **Eval-driven quality.** Tier-wise hallucination control framework adopted by Edge Copilot and Office Copilot; **#1 on Google Natural Questions** for ∼6 months; entity-linking precision from **40% to 90%+** via establishing context-aware personalized eval framework; systematic SBS quality gates for every release.
- **Cross-functional delivery at scale.** Jira Search V-team scaled from ∼30 to **85+ contributors across 8 partner orgs** with a Coalition Council; co-owned Bing Question Answering NRT inference pipeline with **8 platform teams across time zones**; initiated and established **Generative UI Layer — Lumina** across Atlassian 1P apps.
- **Research depth that ships. CodeBERT** (3K+ citations); **Unicoder** (pioneering multilingual pre-trained encoder); Best Paper Award, IJCAI '24 AI4Research; **2× MLADS Distinguished Contribution Award** (first-ever back-to-back); 50+ publications, 11K+ citations, 20+ patents.

Experience

Atlassian (US) — **Feb 2025 – Present**
Head of Machine Learning Engineering — Knowledge AI — *Bellevue, WA*

- **Jira Search rebuild.** MVP in 4 months, full launch in 12; **Search MAU 2.5M → 8.5M (3.4×)**; exceeded AI Org Search MAU L1 OKR. V-team: ∼30 → **85+ contributors across 8 partner orgs**, extended org of **115+**.
- **Enterprise Generative AI Search, 0→GA.** Identified a cross-product gap no team was chartered to own; POC in **<1 month**, GA early 2026. Built **Lumina —** a generative UI framework delivering **20% engagement uplift, 30% SBS lift** vs. competitor AI experiences; adopted as Atlassian's standard GenAI presentation layer. Live demo at **Team '26 Founder Keynote**.
- **Agentic Workflows.** Launched **Jira Agents in Rovo Chat** in 3 months — primary driver of Rovo adoption; code-driven agent architecture beat Manus and GenSpark on **GAIA benchmark**. Built **X-SuperAgent** (Atlassian's first OpenClaw-compliant agent framework), **40%+** developer-productivity uplift.
- **Generative Answers & AI Definitions.** High-utility generative content and novel generative UI drove sustained **AI MAU from 100K to 1.2M**.
- **Knowledge Fundamentals.** Built core knowledge infrastructure (collaboration graph, entity linking, entity recommendation) now adopted by multiple product teams for personalized features. Led entity-linking turnaround from **40% → 90%+ precision —** invested 2 months in hands-on diagnosis before acting, protected team stability through parallel senior hiring, then redirected technical approach.

- **Org leadership.** Extended org of **115+**; grew direct org **30** → **60+** in one year; recruited 2 Principal Managers and 3 Principal Tech Leads; established greenfield sites in Canada and Singapore.

Microsoft — Search Technology Center Asia (Bing, Edge, Copilot) **Aug 2013 – Dec 2024**
Group Senior Principal Applied Sciences Manager (2019–2024) · Senior Dev Manager (2018–2019) · Senior Tech Lead (2016–2018) · SWE II (2013–2015) *Beijing / Suzhou*

- **Built and led an org of 40+ from scratch** (70% scientists, 30% engineers; 5 managers) across Beijing and Suzhou, spanning Bing Search, Copilot, Cross-Lingual QA, Knowledge Graph, Recommendation, and Edge browser AI.
- **Copilot Search (2024–2025).** Multi-turn AI search vertical unifying GenQnA, GenSERP, and Magazine; LLM generation pipelines and R1-distilled task-specific SLMs. **Shipped April 2025 — one month before Google's AI Mode.**
- **Generative SERP & Glow SLM framework (2023–2024).** Built **Glow** — distilled SLMs replacing GPT-4-class models; first-token latency **20–30 s** → ∼**5 ms**; scaled to 5–10% Bing query coverage. Designed tier-wise hallucination control framework for generative search results; subsequently adopted by Edge Copilot and Office Copilot.
- **Generative Answers on Bing (2023–2024). 20% query coverage globally, +2.0% SBS** via streaming generative answers with click-through to Bing Chat.
- **Microsoft Copilot on Edge (2023–2024).** Long-document summarization and chat — **1000+ page docs at 95%+ accuracy**; featured at **Microsoft Copilot 2023 flagship launch event**; shipped to Edge and Windows Copilot.
- **Bing QnA: zero-to-global (2016–2024).** Built from scratch; led evolution through ML → DL → pre-trained models → LLM. Scaled to **100+ languages, 230+ markets**. Authored BERT compression powering Bing's first BERT deployment (within 1 month of open-source, zero GPU cost increase) — **+3.0% coverage, +5.0% precision**. Co-owned NRT inference pipeline with 8 platform teams. **2× MLADS Distinguished Contribution Award**.
- **Query-Dependent Passage Extraction (2019–2022).** Replaced sliding-window indexing with runtime extraction via GNN + multi-teacher distillation (40 ms latency). **#1 on Google Natural Questions for ∼6 months**; adopted across Bing QnA, Knowledge, PAA, and Azure Custom QnA.
- **Knowledge Card (2021–2023).** Multi-modal entity aggregation for 150M entities. **60% Bing query traffic, 65M DAU, +3.0% SBS**; certified as Bing "WoW" feature.
- **NLU Platform "Orca" (2020–2025).** Unified query understanding powering **95%+ of Bing answers**; 60% of onboarded answers closed gap with Google.
- **Segment Entity Search & Recommendation (2021–2024).** Recipe, job, and movie answers — beat Google on relevance/SBS on recipe and job.
- **Underside on Edge (2022–2024).** AI-powered right-rail insight pane. **Edge engaged DAU +10%, Search DAU +1.2M.**

Platforms & Internal Tooling Built

- **Lumina** (Atlassian, 2025–2026) — generative UI framework powering Enterprise GenAI Search; **20% engagement uplift, 30% SBS lift**; adopted as Atlassian's standard GenAI presentation layer.
- **X-SuperAgent** (Atlassian, 2025–2026) — first OpenClaw-compliant agent framework at Atlassian; **40%+** developer-productivity uplift.
- **Glow** (Microsoft, 2023–2024) — distilled SLM serving framework replacing GPT-4-class models; first-token latency **20–30 s** → ∼**5 ms**; scaled to 5–10% Bing query coverage.
- **Orca** (Microsoft, 2020–2025) — unified NLU platform powering answer triggering for **95%+ of Bing answers**.
- **NeuronBlocks** (Microsoft, 2019, open source) — "Build NLP DNN models like playing Lego." Published at EMNLP/IJCNLP 2019.

People & Org Development

- **Founder & Head, Microsoft AI School — Asia-Pacific R&D Group** (2017–2024). Led a **50+ operation v-team**; **10K+ participants, 2K+ honored graduates**; many grew into AI talents and leaders across Microsoft APRD.
- **Invited instructor, LinkedIn Global Academy for Chinese Programmers** (2022–2023): "Intelligent QA Systems for Search Engines" — **50M impressions, 98% satisfaction**.
- **Invited speaker, InfoQ AICon Global AI Conference** (2021): represented Microsoft China on globalization of Bing QA.

Education

Ph.D., Graphics and Visual Computing 2008 – 2013
Institute of Computing Technology, Chinese Academy of Sciences

Selected Awards

- **Best Paper Award**, IJCAI '24 AI4Research — *"Step-Back Profiling: Distilling User History for Personalized Scientific Writing."*
- **2021 Fall Microsoft MLADS Distinguished Contribution Award** (2 / 250+). Broke conference history as first-ever back-to-back recipient.
- **2021 Spring Microsoft MLADS Distinguished Contribution Award** (2 / 300+).
- 2023 Microsoft Global Hackathon — Edge Theatre: **Best Overall Hack**; Greater China Region: **Most Popular Hack**.
- 2023 Microsoft Global Hackathon — WebXT China, DeKG: **Best Innovativeness & Originality Hack**.
- Microsoft AI Core Q3 FY18 **Greatness Award**.
- Excellent Operation Award of AI School China — 2018, 2019, 2020, 2021, 2022, 2023.

Selected Publications (50+ papers, 11K+ citations — scholar profile)

▷ Code Intelligence

- Zhangyin Feng, Daya Guo, Duyu Tang, Nan Duan, Xiaocheng Feng, **Ming Gong**, Linjun Shou, Bing Qin, Ting Liu, Daxin Jiang, Ming Zhou. "CodeBERT: A Pre-Trained Model for Programming and Natural Languages." *EMNLP Findings* 2020. (3,000+ citations)
- Junjie Huang, Duyu Tang, Linjun Shou, **Ming Gong**, et al. "CoSQA: 20,000+ Web Queries for Code Search and Question Answering." *ACL* 2021.

▷ LLM, Agentic AI & Multimodal Pre-training

- Yaobo Liang, Chenfei Wu, Ting Song, et al., **Ming Gong**, Nan Duan. "TaskMatrix.AI: Completing Tasks by Connecting Foundation Models with Millions of APIs." *Intelligent Computing* 2023.
- Gen Li, Nan Duan, Yuejian Fang, **Ming Gong**, Daxin Jiang. "Unicoder-VL: A Universal Encoder for Vision and Language by Cross-modal Pre-training." *AAAI* 2019.
- Nuo Chen, Ning Wu, Shining Liang, **Ming Gong**, et al. "Is Bigger and Deeper Always Better? Probing LLaMA Across Scales and Layers." *arXiv* 2023.
- Ning Wu, **Ming Gong**, Linjun Shou, et al. "Large Language Models are Diverse Role-Players for Summarization Evaluation." *NLPCC* 2023.

▷ Model Training, Compression & Distillation

- **Ming Gong**, Linjun Shou, Wutao Lin, et al. "NeuronBlocks — Building Your NLP DNN Models Like Playing Lego." *EMNLP/IJCNLP* 2019.
- Ze Yang, Linjun Shou, **Ming Gong**, Wutao Lin, Daxin Jiang. "Model Compression with Two-stage Multi-teacher Knowledge Distillation for Web Question Answering System." *WSDM* 2020.
- Fei Yuan, Linjun Shou, Jian Pei, Wutao Lin, **Ming Gong**, Daxin Jiang. "Reinforced Multi-Teacher Selection for Knowledge Distillation." *AAAI* 2021.

▷ Cross-Lingual & Multilingual Modeling

- H. Huang, Yaobo Liang, N. Duan, **Ming Gong**, et al. "Unicoder: A Universal Language Encoder by Pre-training with Multiple Cross-lingual Tasks." *EMNLP/IJCNLP* 2019.
- Nuo Chen, Linjun Shou, **Ming Gong**, Jian Pei, Daxin Jiang. "Bridging the Gap between Language Models and Cross-Lingual Sequence Labeling." *NAACL* 2022.
- Nuo Chen, Linjun Shou, **Ming Gong**, Jian Pei. "From Good to Best: Two-Stage Training for Cross-lingual Machine Reading Comprehension." *AAAI* 2022.
- Shining Liang, **Ming Gong**, Jian Pei, Linjun Shou, Daxin Jiang. "Reinforced Iterative Knowledge Distillation for Cross-Lingual Named Entity Recognition." *KDD* 2021.
- Nuo Chen, Linjun Shou, Tengtao Song, **Ming Gong**, et al. "Structural Contrastive Pretraining for Cross-Lingual Comprehension." *ACL* 2023.
- Linjun Shou, **Ming Gong**, Jian Pei, et al. "Language Scaling: Applications, Challenges and Approaches." *KDD Tutorial* 2021.

▷ Dense Retrieval & RAG

- Shengyao Zhuang, Linjun Shou, Jian Pei, **Ming Gong**, et al. "Typos-aware Bottlenecked Pre-Training for Robust Dense Retrieval." *SIGIR* 2023.
- Houxing Ren, Linjun Shou, Jian Pei, Ning Wu, **Ming Gong**, Daxin Jiang. "Lexicon-Enhanced Self-Supervised Training for Multilingual Dense Retrieval." *EMNLP* 2023.
- Shunyu Zhang, Yaobo Liang, **Ming Gong**, Daxin Jiang, Nan Duan. "Modeling Sequential Sentence Relation to Improve Cross-lingual Dense Retrieval." *ICLR* 2023.

- Shunyu Zhang, Yaobo Liang, **Ming Gong**, Daxin Jiang, Nan Duan. "Multi-View Document Representation Learning for Open-Domain Dense Retrieval." *ACL* 2022.
- Linjun Shou, S. Bo, Fei-Xiang Cheng, **Ming Gong**, et al. "Mining Implicit Relevance Feedback from User Behavior for Web Question Answering." *KDD* 2020.

▷ **Knowledge & Entity Understanding**

- Zilin Xiao, **Ming Gong**, Jie Wu, et al. "Instructed Language Models with Retrievers Are Powerful Entity Linkers." *EMNLP* 2023.
- Zilin Xiao, Linjun Shou, Xingyao Zhang, Jie Wu, **Ming Gong**, et al. "Coherent Entity Disambiguation via Modeling Topic and Categorical Dependency." *EMNLP Findings* 2023.

▷ **Domain-Specific LLMs & Personalization**

- Hanshuang Tong, Jun Li, Ning Wu, **Ming Gong**, et al. "Ploutos: Towards Interpretable Stock Movement Prediction with Financial Large Language Model." *arXiv* 2024.
- (*Best Paper, IJCAI '24 AI4Research*) "Step-Back Profiling: Distilling User History for Personalized Scientific Writing."
- Ning Wu, **Ming Gong**, Linjun Shou, Jian Pei, Daxin Jiang. "RUEL: Retrieval-Augmented User Representation with Edge Browser Logs for Sequential Recommendation." *CIKM* 2023.

Selected Patents (20+ granted; selection below)

- Contrastive Learning for Question Answering. 2020. (MS#407988-CN-NP)
- Providing QA Training Data and Training a QA Model based on Implicit Relevance Feedbacks. 2020. (MS#407927-CN-NP)
- Question Generation from Queries. 2021. (MS#410241-CN-NP)
- Assertion-based Question Answering with Question-Aware Open Information Extraction. 2019.
- Representation Learning of Semi-structured Data. 2020. (MS#408908-CN-NP)
- Graph based Fact Clustering for Diversity and Freshness. 2021.
- Model Generation based Model Compression. 2019. (MS#406805-CN-NP)
- Model Selection Learning for Knowledge Distillation. 2020. (MS#408426-CN-NP)
- Multi-Model Joint Denoising Training. 2021. (MS#409566-CN-NP)
- Lexicon-Enhanced Self-Supervised Training for Multilingual Dense Retrieval. 2022. (MS#412496-CN-NP)
- Personalized Multilingual Content Recommendation based on Robust Feature Network. 2023. (MS#412659-CN-NP)
- Representation Learning of Cross-Language Texts. 2021. (MS#409565-CN-NP)
- Cross-Lingual Task Training. 2019. (MS#406497-CN-NP)
- From Good to Best: Two-Stage Training for Cross-lingual Machine Reading Comprehension. 2021.
- Controversial Poll Auto Generation by Generation Model and Crowdsourcing. 2021.
- Sentence Representation Generation for Cross-lingual Retrieval. 2022.
- Sequential Recommendation based on Cross-Domain Behavior Data. 2023.
- Sequential Recommendation Based on Dual Contrastive Interest Learning. 2022.
- Temporal Co-Contrastive Learning-Based Node Representation Generation. 2022.
- Content Recommendation Based on Graph Enhanced Collaborative Filtering. 2022.
- Knowledge-enhanced Content-Aware Collaborative Filtering for Recommendation. 2021.
- Hierarchical Representation Learning of User Interest. 2021. (MS#410462-CN-NP)